



CAPSTONE PROJECT

Credit Card Default Prediction

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Introduction

Credit card default happens when a customer stops paying what they owe. This project uses real repayment, billing, and demographic data from **30,000 credit card clients in Taiwan (2005)** to predict, before it happens, which customers are likely to default on their next payment.

30,000

customer records

34

features after engineering

6 mo.

of repayment history per client

0

missing values

A binary classification problem: will this client default next month — yes or no?



Problem Statement + Aim

THE PROBLEM

Can we classify customers as likely to default or not, using repayment history, credit limits, bill amounts, and demographics — before a missed payment happens?

THE AIM

Support financial institutions in flagging higher-risk customers early — for risk monitoring and collection strategy — prioritizing Recall, ROC-AUC, and PR-AUC over raw accuracy, since missing an actual defaulter is costlier than a false alarm.



The Data

Each record represents one credit card customer and their 6-month history



CREDIT LIMIT

How much the client can borrow



DEMOGRAPHICS

Age, sex, education, marital status



PAYMENT STATUS

6 months — were they late, and by how long?



BILLS & PAYMENTS

6 months — what they owed vs. what they paid

30,000 records • 25 columns • 0 missing values • Target: default.payment.next.month (Yes / No) • 77.9% no default / 22.1% default



EDA — Key Findings

What the data showed before any modeling began



Repayment status is the top predictor

Customers behind on their most recent payment default far more often than any other group.



Credit limit is protective

Customers who default carry noticeably lower average credit limits than those who don't (\$130K vs. \$178K).



Different balancing methods were tested

Resampling with SMOTENC was tested, but it did not improve the results enough



Outliers were kept, not removed

Unusually high bill amounts and credit limits reflect real high-spending customers, not errors — dropping them would discard real behavior.



Demographics matter less

Education and sex show mild associations with default; marital status shows almost none by comparison.



Modeling Approach



FEATURE ENGINEERING

5 new features over 6-month behavior: DELAY_COUNT, MAX_DELAY, average bill/payment amount, utilization ratio



PREPROCESSING

One-hot encoding has been used



IMBALANCE HANDLING

`class_weight = 'balanced'`



MODELS TESTED

Logistic Regression, Random Forest, and XGBoost — each tuned from a base version with GridSearchCV

34 total features → trained on 24,000 clients → tested on 6,000 unseen clients • Logistic Regression tested as interpretable baseline



External Data Enrichment

Testing whether Taiwan 2005 socioeconomic data adds predictive value

WHAT WAS ADDED

- **Expected monthly income** — based on age group and education
- **Credit-to-income ratio** = Credit limit ÷ Expected monthly income
- **Bill-to-income ratio** = Average monthly bill ÷ Expected monthly income
- **Payment-to-income ratio** = Average monthly payment ÷ Expected monthly income

RESULT — EVALUATED ON RECALL

0.6483

Recall — original features

0.6500

Recall — with external data

External features produced a small improvement in Recall and were retained as supplementary predictors.



Model 1: Logistic Regression

THE INTERPRETABLE BASELINE

TUNING APPROACH

GridSearchCV · refit on Recall

WHAT IMPROVED

Weakest tuned model overall, but by far the most interpretable — every coefficient has a clear sign and story, useful for explaining decisions to non-technical stakeholders.

PERFORMANCE VS. BASE MODEL

Metric	Base	Tuned	Change
Recall	0.297	0.628	+0.331
ROC-AUC	0.760	0.761	+0.001
PR-AUC	0.514	0.512	-0.002



Model 2: Random Forest

THE ENSEMBLE

TUNING APPROACH

GridSearchCV · refit on Recall

WHAT IMPROVED

Random Forest achieved strong ROC-AUC and PR-AUC performance, but its Recall remained below XGBoost.

PERFORMANCE VS. BASE MODEL

Metric	Base	Tuned	Change
Recall	0.372	0.621	+0.249
ROC-AUC	0.770	0.786	+0.016
PR-AUC	0.538	0.562	+0.024



Model 3: XGBoost

THE FINAL PICK ★ BEST

TUNING APPROACH

GridSearchCV · refit on Recall

WHAT IMPROVED

XGBoost achieved the highest cross-validated Recall, ROC-AUC and PR-AUC among the three tuned models and was selected as the final model.

PERFORMANCE VS. BASE MODEL

Metric	Base	Tuned	Change
Recall	0.367	0.652	+0.285
ROC-AUC	0.786	0.787	+0.001
PR-AUC	0.563	0.564	+0.001



Model Comparison

Model	Recall	ROC-AUC	PR-AUC
XGBoost ★	0.652	0.788	0.564
Random Forest	0.621	0.786	0.562
Logistic Regression	0.628	0.761	0.512

WHY XGBOOST

- Highest cross-validated Recall (0.652), ROC-AUC (0.788), and PR-AUC (0.564) among the three models.
- Random Forest was competitive on ROC-AUC and PR-AUC, while Logistic Regression provided a transparent and interpretable baseline.



Conclusion



BUSINESS IMPLICATIONS

- Flags high-risk customers for earlier risk-management action
- Supports credit-limit and collections decisions with data, not guesswork



LIMITATIONS

- Dataset reflects Taiwan, 2005 — may not generalize to other markets or eras
- External features are group-level estimates, not individual income data



FUTURE IMPROVEMENTS

- Use newer customer financial data
- Test the model on customers from different markets and time periods



Thank You

Questions & Discussion

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